ASTR302 spring 2026 Observing at the 61" or 90" Telescopes

The following information is provided to help you plan an observing projects using the Mont4k CCD camera on the 61" telescope on Mt Bigelow, OR, this semester there is the alternative possibility of shadowing other Steward astronomers at the 90" telescope on Kitt Peak, who will either be observing Cepheids in the M33 galaxy with the 90Prime camera or obtaining spectra of new supernovae with the B&C spectrograph, coordinated with Dr. Peter Milne.

Different projects require observations with different instruments, and on different dates or phases of the moon, due to target availability, moon brightness, and other considerations, so it's best to divide yourselves into groups first. Next decide whether you could , choose a project topic and pick appropriate target(s) before deciding which of the ASTR302 observing nights you will go up to the telescope. It's important for each group to try to obtain their data as early in the semester as possible, so there will be enough time for you to try again on a later night if there is bad weather during your first night.

Dr. Green, Dr. Miles and Dr. Milne will provide advice about selecting a suitable topic for your observing project and help you plan the observations that will be needed. The sooner you can go to the telescope for your observing night, the better - if your first assigned night is cloudy, there will be more time for you to try again later.

If you choose a project that can best be done at the 61", your TA, Yu-Hsiu Huang, will drive you up to the telescope, where Dr. Green will give you an introduction to the telescope, supervise your observing and help you obtain your data. The ASTR302 class has been allocated the 61" nights of Feb 20-22 (dark nights), Mar 6-8 (bright nights), Mar 13-15 (dark) and Mar 20-22 (dark). All the 61" nights are on Friday, Saturday or Sunday. If you already have your own idea for a project topic that requires imaging data, we will help you decide if it's feasible to obtain the necessary observations using the 61" Mont4k camera. We can also suggest ideas for project topics that have worked well in the past. Given the uncertain nature of the weather and the limited nights we have at the telescope, it's much better to choose a project that can be accomplished without requiring photometric (ie totally clear) skies and which doesn't need superb seeing conditions. The most versatile projects for the 61"/Mont4k are those in which good observations can be obtained even if the sky is not completely clear. For example, using relative photometry, eg light curves or photometry of a target star relative to the brightness of other stars within the same image, or observing objects that have known secondary standard stars in the same field as the target, instead of having to do absolute photometry relative to standard stars located elsewhere in the sky. In past semesters, successful 61" imaging projects have obtained light curve observations of transiting exoplanets, eclipsing binary stars, pulsating stars, and rotating asteroids, or UBVR imaging of supernovae or Cepheid stars in external galaxies.

If you choose a project where you will participate in ongoing projects at Kitt Peak, you will be driven up to the mountain by Dr. Peter Milne, hopefully on a night that will be clear enough to actually observe. Dr. Sand will observe Cepheids in M33 on the nights of Feb 16 and 17 (Monday and Tuesday; dark nights). The AzTEC supernova group will observe supernova spectra on Feb 9-10 (Monday-Tuesday; gray nights) and Mar 9-14 (Monday through Saturday; gray to dark nights). You would need to work with these observers for advice about the data reduction and to learn about the background science for these projects.

Note: if your project involves UBVR imaging or spectroscopy of supernova, you will have to wait until shortly before your observing night before you can find out whether there is a suitable new supernova target. There is usually a reasonable target to observe, but it's not always possible to predict in advance when a new SN will appear.

Although the 61" is a small telescope, its optics are very good. The Mont4k camera also has an excellent CCD. You can observe point source targets with high S/N as faint as 15th mag, and (in good seeing) fainter targets down to 17th or 18th mag at lower S/N. Fainter than approximately V = 15 requires dark time. 14-15 mag targets need grey time, and even then the targets need to be fairly far away from the moon if you want high precision measurements, as required for most light curves. Targets between mags 11 and 13.5 can be observed almost any time, although within 3-4 days of full moon all targets must be at least 1 to 2 hrs of RA away from the moon and > 15° to 25° away in Declination. Point source targets that are brighter than about magnitude 10.5 are usually too bright for precise measurements with the Mont4k at the 61", because they will saturate the CCD unless the exposure times are very short, which results in a lot of scintillation noise.

To help you plan how to observe projects on the 61", start with the MMT Observer's Almanac. It lists the range of times and the corresponding range of RA's when targets can be observed on a given night. Since the MMT telescope is almost due south of Mt Bigelow, the rising and setting times of the sun and moon and the sidereal times listed in the MMT almanac are nearly the same at the MMT and the 61".

The following paragraphs describe the MMT almanac using an example from a couple of years ago, to familiarize you with how to read the almanac. (You will need to use this year's almanac for planning your observations for this class; the link for the 2026 almanac is given at the end of this section.)

http://www.mmto.org/wp-content/uploads/2024/01/almanac_2024.pdf

(copy and paste this url into a new browser tab or window in order to follow the example below)

In the MMT almanac, each line gives MST (Mountain Standard Time) times for the observing night that starts on the listed civil date. This is rather unusual; most astronomical sites use Universal dates and times (UT) instead; UT is the time at the Greenwich Meridian in England. For example, sunset on Sunday, Sept 1, will occur at 06:47 pm MST; the corresponding Universal Time and date will be 01:47 and Sept 2, respectively. 5 pm MST is always 0^{hr} UT on the following day, because midnight on the Greenwich meridian occurs 7 hrs earlier than it does in Arizona (and Arizona doesn't do Daylight Savings time).

You can use the MMT almanac to figure out what the range of observable RA coordinates will be for any given night, as follows:
Weather permitting, we try to start the process of initializing the telescope position, moving to the target, and focusing the telescope on the CCD about 15 minutes before 12° dark. On Sept 1, that would be 7:25 pm. It will be dark enough to start observing most targets by 20 minutes before 18° dark, ie at 7:50 pm = 02:50 UT. By definition, the Sidereal Time (ST) is equal to the RA of whichever objects are crossing the north/south meridian at that moment. From column 6, the RA of a star which is 3 hrs past the meridian at 18° dark on Sept 1 will be 15:33. The RA of stars crossing the meridian at that same time will be 3 hrs greater, so the ST at 18° dark will be 18:33. At the start of observing, 20 minutes before, the ST will be 18:13. A similar calculation can be done for the times at the end of the night. In good weather, one can usually observe for 25 (or even 30) minutes past 18° dark before the sky starts getting significantly brighter. At the end of the night that started on Sept 1, at 25 mins past 18° dark, the MST will be 5:01 am = 12:01 UT. The ST at 18° dark will be 03:01, 3 hrs less than the value in column 8. 25 mins later, the ST will be 03:26. Therefore, **one would expect to be able to observe typical targets suitable for the 61" between 7:50 pm and 5:01 am on the night of Sept 1, ie between 02:50 and 12:01 UT**. **The corresponding sidereal times will be 18:13 and 03:01**. Targets with RA's a few hours smaller than 03:47 can be observed at the beginning of the night, and ones with RA's a few hours larger than 15:04 can be observed at the end of the night, if you don't need to stay on them for a long time.

The link to the 2026 MMT almanac that you will use to plan your observations for this class is

https://www.mmto.org/wp-content/uploads/2025/11/almanac_2026.pdf

⇨ The UT and ST times, moon phases, etc will be different on different nights; you must check the almanac for each specific night !!

The range of observable Declinations depends on the telescope limits and how low in the sky you are willing to observe your targets. The northern Dec limit is +61°; the telescope cannot point any further north than that due to its English yoke mount. Targets with Dec = –10° can be observed well for no more than 4 to 5 hrs per night because they're so low in the southern sky; targets with even more southerly declinations are observable for even less time each night.

The other very important factor you need to be aware of is the phase and position of the moon – if the moon is too bright or too close to your target, as discussed above, the increase in brightness of the sky background will result in poor (ie noisy) data. The next to last column in the MMT Almanac lists the relative brightness of the moon from full (14+ days) to new (0 days) at midnight on each night. The two previous columns give the time of moonrise and moonset times (MST).

You'll want your target to be more than 25° away from the moon at the very least, ie more than 1 to 2 hrs of RA and 15° to 25° of Dec away, and unless your target is very bright, perhaps a lot more than that. The brighter the moon illumination, the farther away your target should be. For a target with a magnitude about 13.5, when the illumination is > 60%, you should stay at least 2 hrs of RA and 30° Dec away from the moon. For brighter targets or when the moon illumination is lower, it might be okay if the moon is a little closer. However, for fainter targets, you will need to observe very far from the moon, or better, observe when the moon isn't up. Look up the RA and Dec coordinates of the moon by copying and pasting this url into your browser:

http://astropixels.com/ephemeris/moon/moon2025.html

Each line in this website lists the coordinates at 0 hrs on the given UT date (*not* the MST date!). You can see that the moon moves significantly between one night and the next, so make sure the moon won't get too close to your target later in the night!